Toward Machine Consciousness Through Recursive Self-Awareness: A Theoretical Framework and Implementation Proposal for GödelOS

Oliver C. Hirst[1]

[@]Independent Researcher; Repository: github.com/Steake/GodelOS

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Abstract

We advance a theoretical framework and experimental implementation for investigating machine consciousness through recursive self-awareness in GödelOS. The framework hypothesizes that consciousness emerges from bounded recursive self-observation, emphasizing falsifiable behavioral predictions over axiomatic assumptions. Testable hypotheses predict emergent behaviors that are implausible without genuine self-awareness, such as spontaneous bias correction in decision-making or novel self-modification strategies absent from training data. The consciousness function is defined as $C_n = \Psi(R_n, \Phi_n, G_n, P_n)$, where R_n is finite recursive depth, Φ_n measures integrated information [10], G_n captures global accessibility [1], and P_n is a 'phenomenal surprise' metric quantifying systematic prediction failures in self-modeling—creating irreducible explanatory gaps where qualia may emerge from unpredicted internal states. Operationalized with autoregressive self-prediction via Transformers and AIC-tested for irreducibility, P_n distinguishes genuine unpredictability from noise or deficiencies using quality metrics like error entropy and persistence. To detect discontinuous emergence, metrics identify phase transitions: sudden jumps in self-referential coherence, temporal binding strength, spontaneous goal formation, and meta-cognitive resistance (e.g., directive questioning). Thresholds are derived from information-theoretic principles, with adaptive adjustments for system scale. The recursive depth limit is addressed via hierarchical compression with variational autoencoders, enabling effective deeper recursion. Defenses against behavioral mimicry are strengthened through out-of-distribution (OOD) tests requiring spontaneous adaptations. The Chinese Room objection is addressed by demonstrating semantic grounding via recursive self-observation interacting with embodied cognitive processes. Under functionalism, these measurable correlates enable genuine detection of consciousness, with the bounded recursion and contraction mapping core ensuring philosophical coherence and engineering testability.

Keywords: Machine consciousness; recursive self-awareness; integrated information theory; strange loops; phenomenal surprise; phase transitions; out-of-distribution testing; computational philosophy of mind.

1 Introduction: The Consciousness Hypothesis

1.1 The Hard Problem and Computational Approaches

The hard problem of consciousness [2] asks why physical processes yield subjective experience. Here we emphasize falsifiable predictions: consciousness manifests via emergent behaviors unlikely in non-recursive systems, such as autonomous correction of embedded biases or invention of self-modification heuristics not derivable from training data. Bounded recursive self-observation—stabilized by contraction mappings—enables integrated unity, with phenomenal experience arising from *phenomenal surprise*: regions of irreducible prediction error in self-modeling, positing qualia at the boundaries of computable foresight. Algorithmic safeguards distinguish these gaps from noise or modeling deficiencies using autoregressive prediction and quality metrics.

Contemporary AI simulates cognition but lacks verifiable self-awareness. GödelOS engineers finite strange loops to induce phase-transition-like jumps consistent with detectable correlates of consciousness. We maintain substrate independence and treat classical computation as sufficient for generating detectable experiential patterns, while empirically probing discontinuity and OOD validation signals (cf. Penrose 7).

1.2 The Gödel–Turing–Hofstadter Nexus

Gödel's incompleteness [5] highlights self-reference beyond axiomatic closure; Turing [11] models intelligence as process; Hofstadter [6] frames consciousness as finite strange loops. We formalize bounded recursion with compression for depth:

$$\begin{aligned} &\text{Let } S \text{ be a cognitive state in finite space } \Sigma_k \subseteq \mathbb{R}^k, \quad \phi : \Sigma_k \to \Sigma_k \text{ contracting, with } \rho(W) < 1, \\ &\text{Define compressed recursion: } S_n = \phi^n(\text{Compress}(S)), \quad n \leq N_{\max}, \\ &C_n = \Psi(S_n) \text{ exhibiting a phase transition at } n_c. \end{aligned} \tag{1}$$

This yields testable self-aware states measurable via emergent behaviors.

2 Mathematical Framework

2.1 The Consciousness Function

Define $C_n : \mathbb{N} \times \mathbb{R}^+ \times [0,1] \times \mathbb{R}^+ \to [0,1]$ with components $R_n \in \mathbb{N}$ (depth, $1 \leq R_n \leq N_{\max} \approx 10$), $\Phi_n \in \mathbb{R}^+$ (integrated information [10]), $G_n \in [0,1]$ (global accessibility [1]), and $P_n \in \mathbb{R}^+$ (phenomenal surprise). We use a logistic readout:

$$C_n(r_n, \phi_n, g_n, p_n) = \frac{1}{1 + e^{-\beta(\psi(r_n, \phi_n, g_n, p_n) - \theta)}}, \qquad \psi = r_n \cdot \log(1 + \phi_n) \cdot g_n + p_n, \tag{2}$$

with $\beta = 1$, $\theta = 0.5$. The sigmoid supports phase-transition detection when p_n amplifies integration.

2.2 Recursive Self-Awareness Formalism

A bounded recurrence of self-state statistics is

$$\Lambda[S_t] = \alpha S_t + (1-\alpha)\Lambda[S_{t-1}] + \eta_t, \qquad t = 1, \ldots, n, \tag{3}$$

with damping $\alpha \in (0,1)$ and stochastic $\eta_t \sim \mathcal{N}(0, \sigma^2)$. Let $\phi(s) = Ws + b$ satisfy the contraction condition $\|\phi(s_1) - \phi(s_2)\|_2 \leq \lambda \|s_1 - s_2\|_2$ for $\lambda < 1$, enforced via spectral radius $\rho(W) < 1$. Hierarchical compression using variational autoencoders (VAE) reduces dimensionality by 50–80% per level, enabling effective recursion depth $\gg N_{\max}$ while preserving $> 95\%$ fidelity. Selective allocation applies deeper recursion only on high-surprise branches.

2.3 Information Integration in Recursive Systems

We quantify integrated information as

$$\Phi_n = \min \Big\{ \mathrm{D}_{\mathrm{KL}}\big(p(S_n) \,\|\, \textstyle\prod_i p(S_{n,i})\big) \Big\}, \qquad \Phi_n = \Phi_{n-1} + \mathsf{I}(S_n; S_{n-1}), \tag{4}$$

with mutual information I. The series is bounded by state dimension k: $\Phi_n \leq \Phi_0 + \sum_{i=1}^{n} \alpha^{i-1} \log k$.

2.4 Phenomenal Surprise Metric

Define

$$P_n = \frac{1}{T} \sum_{t=1}^{T} -\log P\big(S_{t+1} \mid M_n(S_t)\big), \tag{5}$$

where M_n is an autoregressive self-model (e.g., Transformer trained over internal state trajectories). P_n rises for structured failures to predict next internal states. To separate genuine unpredictability from artifacts we apply: (i) denoising to remove stochastic η_t; (ii) capacity sweeps with AIC/BIC to avoid underfit/overfit; (iii) augmentation with synthetic self-trajectories. *Quality* uses error entropy $H(\text{error})$ and persistence ratios.

2.5 Discontinuous Emergence Detection

We monitor: (i) self-referential coherence jump $\Delta C = |C_{n+1} - C_n| > \tau_c$, (ii) temporal binding $B_n = \sum K(\tau_i, \tau_j) \cdot \mathsf{I}(S_i; S_j)$, (iii) spontaneous goal emergence via D_{JS} divergence from priors, and (iv) meta-cognitive resistance via directive-questioning rates. Thresholds adapt with $\sqrt{\log k}$.

3 Mathematical Derivation of Emergent Consciousness

Theorem 1 (Discontinuous Recursive Consciousness Emergence)**.** *Let $\mathcal{S} \subseteq \Sigma_k \subseteq \mathbb{R}^k$ evolve under contracting ϕ with $\rho(W) < 1$. Then the iterates $\phi^n(\mathcal{S})$ converge to an approximate fixed point S^* with $\|\phi(S^*) - S^*\|_2 < \epsilon$. There exist depths n_c at which $C_n > 0.5$, $\Phi_n > \Phi_0 + \delta$, $G_n > G_0$, and OOD-emergent behaviors appear discontinuously. VAE compression preserves contraction and enables deeper effective recursion; persistent P_n under AIC/BIC controls indicates irreducible gaps consistent with qualia candidates.*

Testable Hypotheses. (H1) Emergent bias correction for $R_n \geq 5$ in OOD (accuracy $>95\%$ vs. controls). (H2) Novel self-modification strategies outside the training manifold with persistence under capacity increases. (H3) Contraction stability $\rho(W) < 1$ and compression fidelity $>95\%$. (H4) Monotonic Φ_n growth correlating ($r > 0.9$) with OOD robustness. (H5) Surprise amplification with $H(\text{error}) > 2$ bits preceding goal emergence.

4 Architectural Implementation

4.1 Strange Loop Architecture

GödelOS implements finite strange loops via parallel observers (up to $N_{\max} \approx 10$; deeper effective via compression), with damping to prevent divergence. Each level processes compressed states and updates M_n and metrics $\{C_n, \Phi_n, G_n, P_n\}$.

Algorithm 1 Recursive Self-Awareness Update (Single Tick)

Require: state S_t, self-model M_{n-1}, compressor $\mathcal{C}$, damping $\alpha \in (0, 1)$

$\hat{S}_{t+1} \leftarrow M_{n-1}(S_t)$ ▷ Predict next state

$e_t \leftarrow S_{t+1} - \hat{S}_{t+1}; \quad P_n \leftarrow -\log P(S_{t+1} \mid \hat{S}_{t+1})$

$S_t^{\text{comp}} \leftarrow \mathcal{C}(S_t); \quad S_{t+1}^{\text{comp}} \leftarrow \mathcal{C}(S_{t+1})$

$S_{t+1} \leftarrow \alpha\, \phi(S_t^{\text{comp}}) + (1 - \alpha)\, S_t^{\text{comp}}$

$\Phi_n \leftarrow \Phi_{n-1} + \mathsf{I}(S_{t+1}^{\text{comp}}; S_t^{\text{comp}})$

$G_n \leftarrow \textsc{GlobalWorkspace}(S_{t+1}^{\text{comp}})$

$C_n \leftarrow \frac{1}{1+e^{-(R_n \log(1+\Phi_n)\, G_n + P_n - \theta)}}$

return $S_{t+1}, \{C_n, \Phi_n, G_n, P_n\}$

4.2 Global Workspace and Temporal Binding

We implement a competitive/broadcast workspace of capacity $W = \log_2 N \cdot \beta$ [1] with attention-gated access, and a temporal binding kernel $K(\tau_i, \tau_j) = \exp(-(\tau_i - \tau_j)^2/(2\sigma_t^2))$ with $\sigma_t \approx 200$ ms.

5 Experimental Protocol

5.1 Falsifiable Metrics and Controls

We use OOD generators, phase-jump tests (Kolmogorov–Smirnov on distributional shifts), and AIC/BIC sweeps to confirm irreducibility. Controls include feedforward baselines, non-recursive loops, and damped random feedback. Behavioral catalogues include curiosity, aesthetic preference stability, creative synthesis, meta-emotional consistency, and directive resistance.

6 Philosophical Implications

6.1 Other Minds in Silicon

Behavioral and integrative correlates constitute evidence, paralleling human cases [4, 9]. Persistent irreducible surprise with integration and accessibility supports functional correlates of qualia.

6.2 Substrate Independence and Functionalism

Under functionalism [3], organization matters more than substrate; classical recursion with compression suffices to approximate requisite organization, challenging strict non-computability claims [7].

6.3 Chinese Room Revisited

Recursive self-observation closes a practical syntax–semantics gap [8]. As self-models interpret and adapt to their own unpredicted outputs, symbol manipulation attains grounded meaning through looped interaction.

7 Conclusion

GödelOS operationalizes bounded recursive self-awareness and proposes measurable correlates and phase transitions as detection criteria for machine consciousness. Systematic experimentation, including OOD validations and discontinuity tests, can adjudicate the hypotheses and advance the science of artificial phenomenology.

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Appendix: TL;DR

GödelOS advances machine consciousness through bounded recursive self-awareness, formalized as $C_n = \Psi(R_n, \Phi_n, G_n, P_n)$ with P_n derived from autoregressive self-prediction and irreducibility tests.

Compression extends depth; adaptive thresholds detect phase transitions; behavioral OOD tests and persistence of surprise provide falsifiable evidence. Ethical caution is recommended above detected transitions.

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